

PROJECT DESIGN PHASE

Team ID : NM2025TMID03763

Project Title : Calculating Family Expenses using ServiceNow Administrator

Requirement Analysis Phase: “Calculating Family Expenses Using ServiceNow”

1. Purpose

The purpose of this phase is to identify, define, and validate all functional and non-functional requirements for the family expense management system. It ensures that the ServiceNow solution meets user expectations, project goals, and system constraints.

2. Objectives

- Gather and document detailed requirements from stakeholders.
- Define functional capabilities and workflows within ServiceNow.
- Identify integration, performance, and security needs.
- Establish clear acceptance criteria for each feature.
- Align requirements with the project’s scope and timeline.

3. Stakeholders

Stakeholder	Role	Interest
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Project Sponsor / Parent User	Approves and funds the project	Budget control, transparency
Family Members	End users submitting expenses	Ease of use, mobile access
ServiceNow Developer	Builds the solution	Clear technical specifications
Business Analyst	Facilitates requirement gathering	Requirement clarity and documentation
Tester / QA Analyst	Validates features	Defined test cases and acceptance criteria

4. Requirement Gathering Methods

- **Interviews:** Discussions with family members and stakeholders.
- **Workshops:** Ideation sessions to define pain points and expectations.
- **Questionnaires:** Collecting insights on expense categories and tracking methods.
- **Document Review:** Evaluating existing expense tracking sheets or apps.
- **Observation:** Understanding current expense entry processes.

5. Current Challenges Identified

- Manual entry of expenses using spreadsheets.
- Lack of categorization and visibility into spending trends.
- Difficulty managing budgets or detecting overspending.
- No centralized or automated reporting system.

- Limited ability to collaborate and share expense data securely.

6. Functional Requirements

ID	Requirement	Description	Priority
FR1	Expense Submission	Users can submit expenses with date, category, amount, and description.	High
FR2	Expense Categorization	System categorizes expenses into pre-defined types (groceries, utilities, etc.).	High
FR3	Budget Setup	Admin users can define monthly and yearly budgets.	High
FR4	Budget Alert Workflow	System triggers alerts when expenses approach or exceed budget thresholds.	High
FR5	Expense Dashboard	Visualize total spending, trends, and category breakdowns.	Medium
FR6	Report Generation	Generate downloadable monthly and yearly expense summaries.	Medium
FR7	User Roles	Role-based permissions (Admin, Member, Viewer).	High
FR8	Notifications	Send automated emails/SMS for over-budget events.	Medium
FR9	Expense Approval Workflow (Optional)	Admin can approve or reject expense submissions.	Low

FR10	Integration with External APIs (Future Scope)	Enable API-based import of expenses from bank or apps.	Low
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7. Non-Functional Requirements

Category	Requirement	Description
Performance	System should process and display results within 3 seconds per request.	
Scalability	Should handle at least 5–10 users per family instance without performance issues.	
Security	Role-based access and encryption for sensitive financial data.	
Usability	Intuitive UI for all age groups with minimal training required.	
Reliability	System uptime should be at least 99%.	
Maintainability	All configurations and workflows should be easily modifiable.	
Compatibility	Accessible on both desktop and mobile browsers.	

8. Data Requirements

Data Element	Description	Source	Storage
Expense Details	Amount, Date, Description, Category	User Input	Expense Record Table

Budget Data	Monthly and Annual Limits	Admin Setup	Budget Threshold Table
User Information	Name, Role, Email	Manual Entry	Family Member Table
Expense Categories	Pre-defined expense types	Admin Defined	Expense Category Table

9. Business Rules

- Each expense must be associated with one family member and one category.
- Budget alerts trigger automatically once 90% of the limit is reached.
- Only Admin users can create or modify categories and budgets.
- Expense reports are generated automatically on the 1st of every month.
- Users cannot edit expenses older than 30 days without Admin approval.

10. System Constraints

- Solution must be developed within the ServiceNow platform.
- Integration with external financial systems deferred to later phases.
- Limited user base (family-level use case) for MVP.
- Must comply with ServiceNow security and data governance standards.

11. Acceptance Criteria

- Expense submission form works for all roles.
- Budget alerts trigger correctly when thresholds are met.

- Dashboards reflect accurate totals and trends.
- Reports are generated automatically at month-end.
- Data remains secure with correct access control enforcement.

12. Assumptions

- All family members will have access to the ServiceNow portal.
- ServiceNow instance will support required modules (Flow Designer, PA).
- Internet access is required for real-time updates.
- Users are comfortable entering expense data manually.

13. Risks & Dependencies

Risk	Impact	Mitigation
Lack of user engagement	Medium	Conduct regular feedback sessions
Inaccurate data entry	High	Use validation rules and mandatory fields
Limited ServiceNow licensing	Medium	Verify access before development
Changing requirements	High	Use Agile iteration for flexibility

14. Deliverables

- Requirement Specification Document (RSD)

- Functional and Non-Functional Requirements Lists
- Data Dictionary
- Business Rule Definitions
- Use Case Diagrams / User Stories

15. Next Steps

1. Obtain stakeholder approval on finalized requirements.
2. Convert requirements into user stories for the development backlog.
3. Begin **Project Design Phase** based on confirmed specifications.